

Jaehyuk Choi

Ph.D.

Mathematics Hall Rm 426
2990 Broadway
New York, NY 10027
✉ jc6569@columbia.edu

🌐 www.math.columbia.edu/mafn/learn-more/

Research Interest

Mathematical Finance, Financial Technology, Machine Learning, Numerical Methods

Work Experience

- 2025.1 – Present **Mathematics of Finance (MAFN) MA Program Director**, *Columbia University*, New York, US
- 2022.2 – 2024.12 **Tenured Associate Professor**, *Peking University HSBC Business School*, Shenzhen, China
2016.8 – 2022.1 **Assistant Professor**
- Taught core and elective courses for quantitative finance and fintech master program.
 - A faculty member of Sargent Institute of Quantitative Economics and Finance (SIQEF).
- 2023.8 – 2024.8 **Visiting Scholar**, *Department of Mathematics (Financial Engineering)*, *Baruch College, City University of New York*, (Host: Profs. [Jim Gatheral](#) and [Dan Stefanica](#))
- 2022 Summer **Visiting Scholar**, *Department of Mathematics, Hong Kong University of Science and Technology (HKUST)*, (Host: Prof. [Yue Kuen Kwok](#))
- 2019, 2020, 2021 Summer **Visiting Scholar**, *Department of Mathematics, Hong Kong University of Science and Technology (HKUST)*, (Host: Prof. [Lixin Wu](#))
- 2018.2 **Visiting Scholar**, *Department of Mathematics, City University of Hong Kong*, (Host: Prof. [Qingshuo Song](#))
- 2013.4 – 2016.7 **Quantitative Analyst**, *Goldman Sachs Asia*, Hong Kong, China
- Executive Director, Asia Pacific Ex-Japan interest rate vol & exotics desk, FICC
 - Developed interest rate/FX models for trades and maintained pricing/risk system (Slang/SecDB).
 - Led US federal reserve stress test in Asia macro trading division.
- 2007 – 2013 **Quantitative Analyst**, *Goldman Sachs*, New York, US
- Vice President (since 2009), US interest rate volatility & exotics and inflation desk, FICC
 - Developed interest rate models for trading. Maintained pricing/risk system (Slang/SecDB).
- 2005 – 2007 **Quantitative Analyst**, *BNP Paribas*, New York, US
- Associate, US interest rate exotics desk, hired by Alan Brace and Marek Musiela

Education

- 2000 – 2005 **Ph.D., Applied Mathematics**, *Massachusetts Institute of Technology*, Cambridge MA, US
- Advisor: Prof. [Martin Z. Bazant](#)
 - Dissertation: Transport-limited aggregation and dense granular flow
- 2002 – 2004 **Financial Technology Option**, *MIT*, Cambridge MA, US
- Certificate program run by Sloan School & Dept. of EECS, sponsored by Merrill Lynch
- 1994 – 2000 **B.S., Mathematics**, *Korea Advanced Institute of Science Technology (KAIST)*, Daejeon, Korea

Note: * indicates the corresponding author.

■ Working Papers

- [1] Jaehyuk Choi*, Lan Ju, Jian Li, and Zhiyong Tu. Information extraction and artwork pricing, 2023. [arXiv:2302.08167](#).

■ Publications in Quantitative Finance

- [2] Jaehyuk Choi*, Lilian Hu, and Yue Kuen Kwok. Efficient and accurate simulation of the stochastic-alpha-beta-rho model. *European Journal of Operational Research*, 329(1):166–179, 2026. [doi:10.1016/j.ejor.2025.09.027](#).
- [3] Jaehyuk Choi*. Exact simulation scheme for the Ornstein–Uhlenbeck driven stochastic volatility model with the Karhunen–Loève expansions. *Operations Research Letters*, 60:107280, 2025. [doi:10.1016/j.orl.2025.107280](#).
- [4] Jaehyuk Choi*, Jeonggyu Huh, and Nan Su. Tighter ‘uniform bounds for Black–Scholes implied volatility’ and the applications to root-finding. *Operations Research Letters*, 57:107189, 2024. [doi:10.1016/j.orl.2024.107189](#).
- [5] Jaehyuk Choi* and Byoung Ki Seo. Option pricing under the normal stochastic alpha-beta-rho model with Gaussian quadratures. *Journal of Computational Finance*, 28(2):1–20, 2024. [arXiv:2301.02797](#), [doi:10.21314/JCF.2024.007](#).
- [6] Jeechul Woo, Chenru Liu, and Jaehyuk Choi*. Leave-one-out least squares Monte Carlo algorithm for pricing Bermudan options. *Journal of Futures Markets*, 44(8):1404–1428, 2024. [arXiv:1810.02071](#), [doi:10.1002/fut.22515](#).
- [7] Jaehyuk Choi* and Yue Kuen Kwok. Simulation schemes for the Heston model with Poisson conditioning. *European Journal of Operational Research*, 314(1):363–376, 2024. [arXiv:2301.02800](#), [doi:10.1016/j.ejor.2023.10.048](#).
- [8] Jaehyuk Choi, Desheng Ge, Kyu Ho Kang, and Sungbin Sohn. Yield spread selection in predicting recession probabilities. *Journal of Forecasting*, 42(7):1772–1785, 2023. [arXiv:2101.09394](#), [doi:10.1002/for.2980](#).
- [9] Jaehyuk Choi, Lei Lu, Heungju Park*, and Sungbin Sohn. The financial value of the within-government political network: Evidence from Chinese municipal corporate bonds. *Finance Research Letters*, 47:102552, 2022. [arXiv:2201.01160](#), [doi:10.1016/j.frl.2021.102552](#).
- [10] Jaehyuk Choi* and Rong Chen. Improved Iterative Methods for Solving Risk Parity Portfolio. *Journal of Derivatives and Quantitative Studies*, 30(2):114–124, 2022. [Open Access]. [doi:10.1108/JDQS-12-2021-0031](#).
- [11] Jaehyuk Choi*, Minsuk Kwak, Chyng Wen Tee, and Yumeng Wang. A Black-Scholes user’s guide to the Bachelier model. *Journal of Futures Markets*, 42(5):959–980, 2022. [arXiv:2104.08686](#), [doi:10.1002/fut.22315](#).
- [12] Jaehyuk Choi* and Lixin Wu. The equivalent constant-elasticity-of-variance (CEV) volatility of the stochastic-alpha-beta-rho (SABR) model. *Journal of Economic Dynamics and Control*, 128:104143, 2021. [arXiv:1911.13123](#), [doi:10.1016/j.jedc.2021.104143](#).

- [13] Jaehyuk Choi* and Lixin Wu. A note on the option price and ‘Mass at zero in the uncorrelated SABR model and implied volatility asymptotics’. *Quantitative Finance*, 21(7):1083–1086, 2021. [arXiv:arXiv:2011.00557](#), [doi:10.1080/14697688.2021.1876908](#).
- [14] Jaehyuk Choi*, Yeda Du, and Qingshuo Song. Inverse Gaussian quadrature and finite normal-mixture approximation of the generalized hyperbolic distribution. *Journal of Computational and Applied Mathematics*, 388:113302, 2021. [arXiv:1810.01116](#), [doi:j.cam.2020.113302](#).
- [15] Carol Alexander*, Jaehyuk Choi, Hamish R. A. Massie, and Sungbin Sohn. Price discovery and microstructure in ether spot and derivative markets. *International Review of Financial Analysis*, 71:101506, 2020. [[Download](#)]. [doi:10.1016/j.irfa.2020.101506](#).
- [16] Carol Alexander, Jaehyuk Choi, Heungju Park, and Sungbin Sohn*. BitMEX bitcoin derivatives: Price discovery, informational efficiency and hedging effectiveness. *Journal of Futures Markets*, 40(1):23–43, 2020. [[Download](#), A top-cited article in the journal in 2020]. [doi:10.1002/fut.22050](#).
- [17] Jaehyuk Choi*, Chenru Liu, and Byoung Ki Seo. Hyperbolic normal stochastic volatility model. *Journal of Futures Markets*, 39(2):186–204, 2019. [arXiv:arXiv:1809.04035](#), [doi:10.1002/fut.21967](#).
- [18] Jaehyuk Choi. Sum of all Black-Scholes-Merton models: An efficient pricing method for spread, basket, and Asian options. *Journal of Futures Markets*, 38(6):627–644, 2018. [arXiv:1805.03172](#), [doi:10.1002/fut.21909](#).
- [19] Jaehyuk Choi and Sungchan Shin*. Fast swaption pricing in Gaussian term structure models. *Mathematical Finance*, 26(4):962–982, 2016. [arXiv:arXiv:1803.08803](#), [doi:10.1111/mafi.12077](#).
- [20] Jaehyuk Choi, Kwangmoon Kim*, and Minsuk Kwak. Numerical approximation of the implied volatility under arithmetic Brownian motion. *Applied Mathematical Finance*, 16(3):261–268, 2009. [[Download](#)]. [doi:10.1080/13504860802583436](#).

■ Publications in Applied Mathematics and Physics

- [21] Jaehyuk Choi, Darren Crowdy, and Martin Z Bazant. Diffusion-limited aggregation on curved surfaces. *EPL (Europhysics Letters)*, 91(4):46005, 2010. [[Download](#)]. [doi:10.1209/0295-5075/91/46005](#).
- [22] Dionisios Margetis* and Jaehyuk Choi. Generalized iteration method for first-kind integral equations. *Studies in Applied Mathematics*, 117(1):1–25, 2006. [[Download](#)]. [doi:10.1111/j.1467-9590.2006.00340.x](#).
- [23] Benny Davidovitch, Jaehyuk Choi, and Martin Z Bazant. Average shape of transport-limited aggregates. *Physical Review Letters*, 95(7):075504, 2005. [[Download](#)]. [doi:10.1103/PhysRevLett.95.075504](#).
- [24] Jaehyuk Choi, Dionisios Margetis, Todd M Squires, and Martin Z Bazant. Steady advection–diffusion around finite absorbers in two-dimensional potential flows. *Journal of Fluid Mechanics*, 536:155–184, 2005. [[Download](#)]. [doi:10.1017/S0022112005005008](#).
- [25] Jaehyuk Choi, Arshad Kudrolli, and Martin Z Bazant. Velocity profile of granular flows inside silos and hoppers. *Journal of Physics: Condensed Matter*, 17(24):S2533, 2005. [[Download](#)]. [doi:10.1088/0953-8984/17/24/011](#).
- [26] Jaehyuk Choi, Arshad Kudrolli, Rodolfo R Rosales, and Martin Z Bazant. Diffusion and mixing in gravity-driven dense granular flows. *Physical Review Letters*, 92(17):174301, 2004. [[Download](#)]. [doi:10.1103/PhysRevLett.92.174301](#).

- [27] Martin Z Bazant, Jaehyuk Choi, and Benny Davidovitch. Dynamics of conformal maps for a class of non-laplacian growth phenomena. *Physical Review Letters*, 91(4):045503, 2003. [Download]. doi: [10.1103/PhysRevLett.91.045503](https://doi.org/10.1103/PhysRevLett.91.045503).

■ Miscellaneous

- [28] Jaehyuk Choi. Finance, Mathematics and Quants. *Newsletter of the Korean Mathematical Society*, 1(153):1–12, 2014. [In Korean, Download].
- [29] Martin Z Bazant, Jaehyuk Choi, Benny Davidovitch, and Darren Crowdy. Transport-limited aggregation. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 14(4):S6, 2004. doi:[10.1063/1.1821713](https://doi.org/10.1063/1.1821713).
- [30] Martin Z Bazant, Jaehyuk Choi, and Benny Davidovitch. Advection-diffusion-limited aggregation. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 14(4):S7, 2004. doi:[10.1063/1.1821714](https://doi.org/10.1063/1.1821714).

Teaching and Student Advising

■ Courses Taught at Columbia University’s MAFN Program

Fall 2025 **Stochastic Processes – Applications**, (4.11/5.0)

A mandatory course of the MAFN program

■ Master-level Courses Taught at Peking University

Note: Teaching evaluation is in the form of (**my score**/school average).

2016 – 2024 **Stochastic Finance**, (**97.15**/95.13, **98.55**/94.33, **99.15**/93.50, **95.27**/95.12, **98.13**/95.7, **94.79**/95.6, **99.2**/96.36, **95.46**/95.95)

A stochastic calculus course required for quantitative finance program.

2016 – 2024 **Applied Stochastic Processes**, (**97.77**/93.88, **97.60**/93.31, **93.50**/92.47, **94.94**/92.73, **100**/95.7, **99.3**/95.6, **100**/95.68, **99.18**/95.95)

A project-based, application-focused course with Python implementation.

2016 – 2024 **Machine Learning for Finance**, (a.k.a., Topics in Quantitative Finance), (**96.44**/95.13, **97.63**/94.33, **91.07**/92.47, **98.87**/95.12, **98.07**/95.45, **98.63**/96.36, **100**/95.95)

An introductory machine learning course with Python `scikit-learn` package.

2018 **Blockchain and Digital Currency**, (**97.71**/93.50)

A required course for fintech program.

2019 – 2024 **Research Methodology**, (for Quantitative Finance and Fintech), (**98.87**/95, **96.49**/92.73, **95.6**/95.7, **97.38**/95.45, **98.89**/95.68, **99.93**/96.36, **96.59**/95.95)

A course for preparing students for thesis writing.

■ Ph.D. Placement at Peking University (by graduation year)

2020 **Shiqi Zhang**, Sauder School of Business (Finance), University of British Columbia (UBC)

2020 **Linsheng Zhuang**, Operations Research and Analytics, National University of Singapore (NUS)

2018 **Chenru Liu**, Management Science and Engineering, Stanford University

■ Master Students Advising at Peking University (by graduation year)

2024 **6 students**

2023 **6 students**

2022 **4 students**

2021 **9 students**, (The best master thesis award of the year included)

2020 11 students
2019 8 students
2018 4 students

Professional Activity

■ Open-source Financial Analytics

PyFENG: **Python Financial ENGineering**

- PyPI: <https://pypi.org/project/pyfeng>
- Code: <https://github.com/PyFE/PyFENG>
- Documentation (html): <https://pyfeng.readthedocs.io>

FER: **Financial Engineering in R**

- CRAN: <https://cran.r-project.org/package=FER>
- Code: <https://github.com/PyFE/FE-R>
- Documentation (PDF): <https://cran.r-project.org/web/packages/FER/FER.pdf>

■ Journal Referee

Journal of Banking & Finance | Journal of Futures Markets | Finance Research Letters | Financial Innovation | International Review of Financial Analysis | North American Journal of Economics and Finance | Applied Mathematical Finance | Journal of Derivatives and Quantitative Studies | AIMS Mathematics | Korean Journal of Financial Studies | Digital Finance | Decisions in Economics and Finance | Insurance: Mathematics and Economics | European Journal of Operational Research

■ Conference and Seminar Presentations

- 2025 Bloomberg BBQ ([Link](#)) | Utrecht U. | Bayes Business School | The 21th Asia-Pacific Assoc. of Derivatives (APAD) Conf.
- 2024 Pusan National U. | Joint Conf. with the Allied Korea Fin. Assoc. | Korea–Japan Math. Fin. Conf. | UConn | The 12th Bachelier World Congress | The Chinese U. of Hong Kong – SEEM | Columbia U. IEOR | Stevens Inst. of Tech.
- 2023 SIAM–FM ([Link](#)) | WPI ([Link](#)) | INFORMS | Bloomberg BBQ ([Link](#))
- 2022 Seoul National U. | Ajou U.
- 2021 The 5th PKU-NUS Conf. | The 2021 SUSTech Workshop on Fin. Engineering | The 2021 Korea Derivatives Assoc. Annual Conf.
- 2020 2020 Global AI Fin. Research Conf. | The 20th China Econ. Annual Conf.
- 2019 The 4th PKU-NUS Conf. | SIAM–FM ([Link](#)) | The 15th Asia-Pacific Assoc. of Derivatives (APAD) Conf. ([Link](#)) | The 3rd KAFE-JAFEE Intl. Conf. on Fin. Engineering ([Link](#)) | Sookmyung Women’s U. | Quant. Methods in Finance 2019 Conf. ([Link](#))
- 2018 INFORMS Intl. ([Link](#)) | The 14th Asia-Pacific Assoc. of Derivatives (APAD) Conf. ([Link](#)) | The 10th Bachelier World Congress ([Link](#)) | Intl. Conf. on Math. Fin. & Symposium on the Role of Math. Fin. on FinTech Business ([Link](#)) | Chung-Ang U.
- 2017 The 5th Asian Quant. Finance Conf. ([Link](#)) | Yonsei U. ([Link](#)) | The 13th Asia-Pacific Assoc. of Derivatives (APAD) Conf. ([Link](#)) | UNIST ([Link](#)) | INFORMS ([Link](#))
- 2016 HKUST | Seoul National U. ([Link](#)) | Ajou U.

■ Academic Membership

2019 – Korea Derivatives Association

- 2021 – Korean Finance Association, Korean Securities Association, Korea Money and Finance Association
- 2022 – Korean Association of Financial Engineering (Executive Director since 2022)

Awards and Grants

- 2026 **aiX Faculty Fellowship Program**, *Google and Columbia U Data Science Institute*
- 2022, 2018, 2017 **Bridge Trust Asset Management Research Fund**
- Improving weight solving method for risk parity portfolio
 - A PCA method for detecting sudden liquidation of quantitative funds
 - Deep Learning Applied to Finance
- 2016 – Present **Faculty Award of Excellence (2019, 2020, 2021, 2022, 2024)**, *Peking University HSBC Business School*
- 2018 **Best Paper Award**, *The 14th Asia-Pacific Association of Derivatives (APAD) conference*
- 2017 – 2018 **Course Development Grant**, *Peking University Graduate Course Project Construction Fund for “Stochastic Finance”*
- 2004 **Best Images in the Gallery of Statistical and Nonlinear Physics**, *American Physical Society*
- 2000 – 2005 **Scholarship**, *Korea Foundation for Advanced Studies*
- 2000 – 2002 **Department Fellowship**, *MIT Department of Mathematics*
- 1993 **Honorable Mention**, *International Physics Olympiad (IPhO)*, College of William and Mary, Williamsburg VA, US